Separable Differential Equations

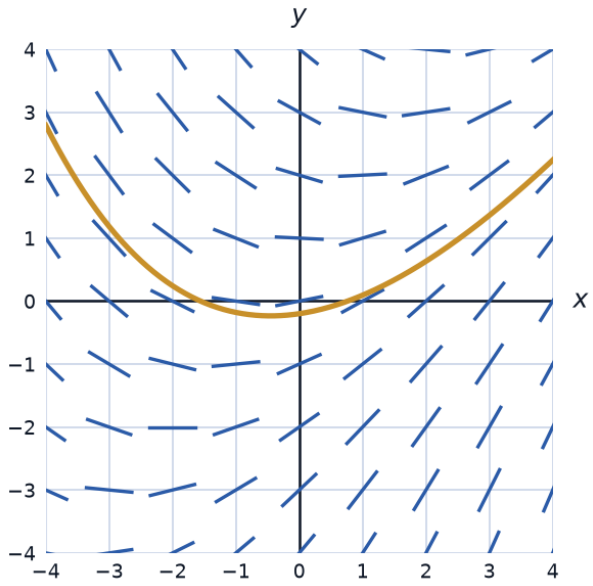
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

2 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.



3 **[Foundational]** • **Target: Growth, decay, and applications**

A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.

a Solve for $P(t)$.

b Find $P(20)$, to the nearest whole number.

4 **[Foundational]** • **Target: Growth, decay, and applications**

Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°C$ room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.

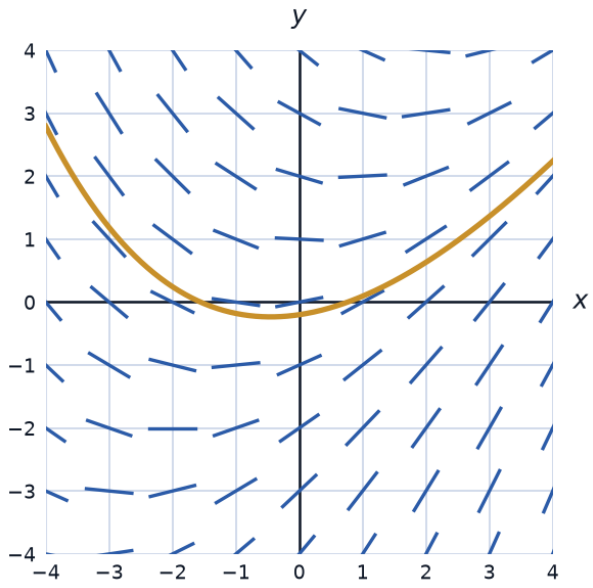
5 **[Foundational]** • **Target: More separable equations**

Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.

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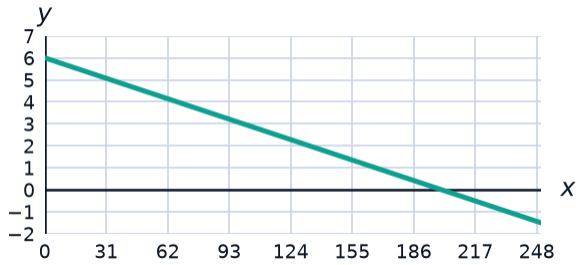
6 [Foundational] • Target: More separable equations

Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.



7 [Foundational] • Target: Mixing and applied models

A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t. The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.



a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$).

b Explain, in words, what this equilibrium represents physically.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

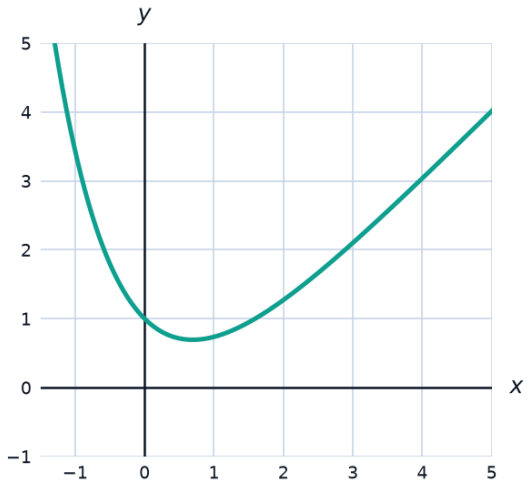
8 [Intermediate] • Target: Slope fields and qualitative behavior

For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.

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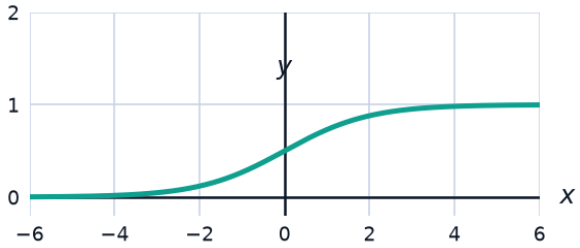
9 [Intermediate] • Target: Slope fields and qualitative behavior

Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.



10 [Intermediate] • Target: Visualizing a logistic solution curve

The graph shows the logistic solution $y = \frac{e^x}{1 + e^x}$ found earlier for $\frac{dy}{dx} = y(1 - y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \to \infty$ and as $x \to -\infty$.



11 [Intermediate] • Target: Another applied separable equation

Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.

12 [Intermediate] • Target: Separable equations with algebraic solutions

Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$, given $y(0) = 2$.

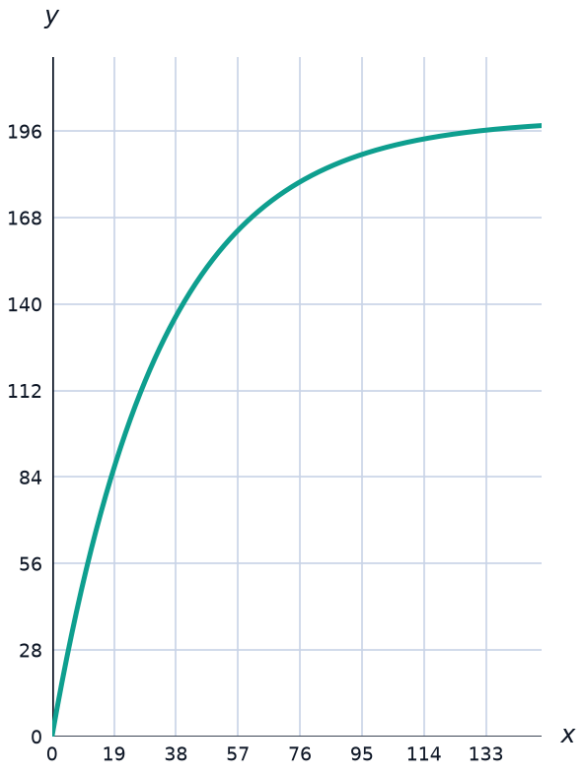
13 [Intermediate] • Target: Separable equations with algebraic solutions

Solve $\frac{dy}{dx} = \frac{1}{2y}$, given $y(0) = 3$.

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14 [Intermediate] • Target: Verifying a solution to an applied model

For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \to \infty$.



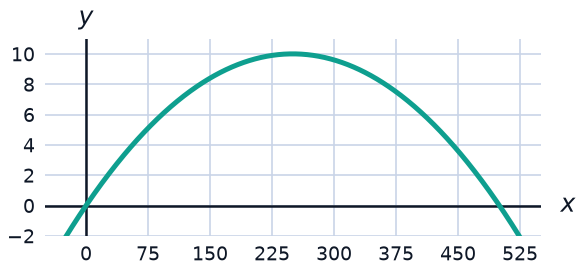
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.

16 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{500}\right)$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.



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17 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 6.01 Related Rates

Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.

18 [Challenging] • Target: A final applied separable model • Cross-Lesson: 6.02 Optimization and Curve Sketching

A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.

19 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 7.01 Integration by Substitution

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:.

20 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 7.02 Integration by Parts

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Solutions of Separable Equations

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$.

a. $y^2 = 2x^2 + C$

b. $y = x^2 + C$

c. $y^2 = x^2 + C$

d. $y = 2x^2 + C$

22 [Foundational] • Target: General Solutions of Separable Equations

Find the general solution to $\frac{dy}{dx} = 3x^2y$.

a. $y = Ce^{x^3}$

b. $y = e^{x^3} + C$

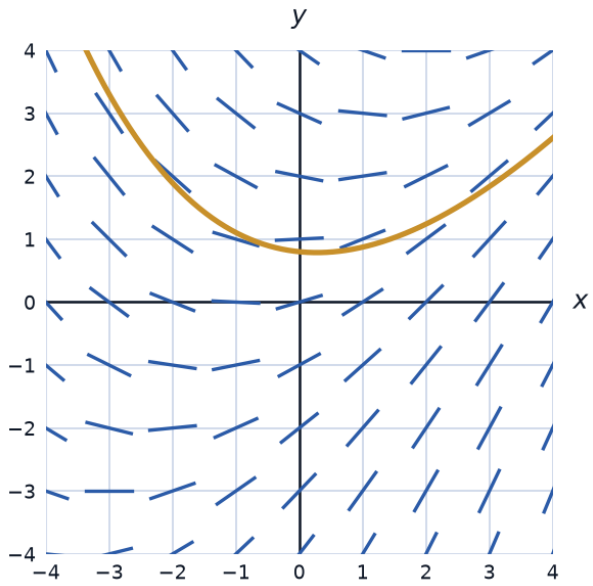
c. $y = Cx^3$

d. $y = 3x^3 + C$

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23 [Foundational] • Target: General Solutions of Separable Equations

Find the general solution of $\frac{dy}{dx} = (1 + y^2)e^x$.



a. $\arctan(y) = e^x + C$

b. $y = \tan(e^x) + C$

c. $y = e^x + C$

d. $\ln(1 + y^2) = e^x + C$

24 [Foundational] • Target: Initial Value Problems (IVPs)

Solve the initial value problem $\frac{dy}{dx} = 2xy$ with $y(0) = 3$.

a. $y = 3e^{x^2}$

b. $y = e^{x^2} + 2$

c. $y = 3e^{2x}$

d. $y = 3x^2 + 3$

25 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dx} = -\frac{x}{y}$ with $y(3) = 4$.

a. $x^2 + y^2 = 25$

b. $x^2 - y^2 = -7$

c. $y = -x + 7$

d. $x^2 + y^2 = 7$

26 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dt} = ky$ given $y(0) = 50$ and $y(2) = 200$. What is the formula for $y(t)$?

a. $y(t) = 50 \cdot 2^t$

b. $y(t) = 50e^{2t}$

c. $y(t) = 50 \cdot 4^t$

d. $y(t) = 50 + 75t$

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27 [Foundational] • Target: Slope Fields and Solution Curves

Along which curve in the xy-plane do the line segments in the slope field for $\frac{dy}{dx} = x - y$ have slope 0?

a. Along the line $y = x$

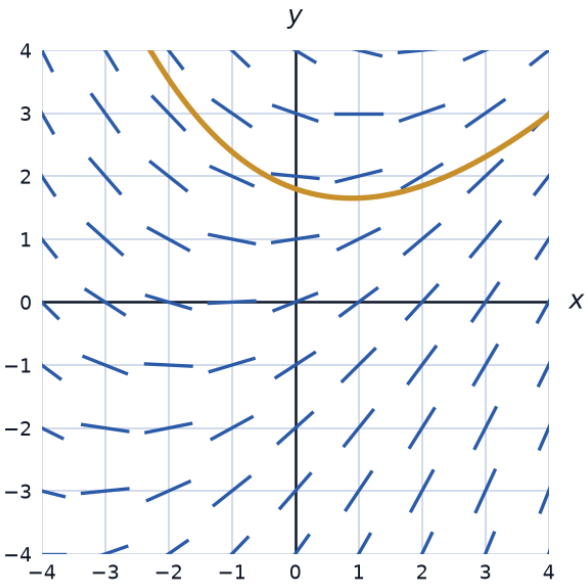
b. Along the x-axis ($y = 0$)

c. Along the y-axis ($x = 0$)

d. Along $y = -x$

28 [Foundational] • Target: Slope Fields and Solution Curves

A slope field shows line segments that depend ONLY on y (horizontal rows of equal slopes). Which differential equation could this represent?



a. $\frac{dy}{dx} = y(2 - y)$

b. $\frac{dy}{dx} = x^2 - y$

c. $\frac{dy}{dx} = xy$

d. $\frac{dy}{dx} = 3x + 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Slope Fields and Solution Curves

For $\frac{dy}{dx} = x^2 + y^2$, what can be said about all non-origin slopes in the slope field?

a. All slopes are strictly positive

b. Slopes are positive in quadrants I and III only

c. Slopes are 0 everywhere

d. Slopes are negative when $x < 0$

30 [Intermediate] • Target: Logistic Growth Models

In the logistic differential equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$, what is the carrying capacity M?

a. 800

b. 0.05

c. 400

d. 16000

31 [Intermediate] • Target: Logistic Growth Models

At what population P is the growth rate $\frac{dP}{dt}$ maximized in the logistic equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$?

a. $P = 400$

b. $P = 800$

c. $P = 200$

d. $P = 0$

32 [Intermediate] • Target: Logistic Growth Models

What is $\lim_{t \to \infty} P(t)$ for any solution to $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$ with $P(0) = 50$?

a. 800

b. ∞

c. 400

d. 50

33 [Intermediate] • Target: Slope Fields and Solution Curves

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$

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34 [Intermediate] • Target: Logistic Growth Models

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

35 [Intermediate] • Target: Solving by separation of variables

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1 e^{r_1 x} + C_2 e^{-r_1 x}$

b. $y = C_1 e^{r_1 x}$

c. $y = (C_1 + C_2) e^{2r_1 x}$

d. $y = (C_1 + C_2 x) e^{r_1 x}$

36 [Intermediate] • Target: Growth, decay, and applications

$y'' + y = 0$: general solution:

a. $y = C_1 \sin x + C_2 \cos x$

b. $y = C_1 e^x + C_2 e^{-x}$

c. $y = C_1 x + C_2$

d. $y = C_1 e^{-x}$

37 [Intermediate] • Target: More separable equations

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$

d. $-\cos x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mixing and applied models • Cross-Lesson: 6.02 Optimization and Curve Sketching

First-order linear ODE solution structure: general solution = homogeneous solution +

a. particular solution

b. another homogeneous solution

c. the initial condition

d. zero

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39 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 7.01 Integration by Substitution

Logistic model: $dP/dt = 0.1P(1 - P/100)$, $P(0) = 10$. $P \to$ as $t \to \infty$:

a. 0

b. 10

c. ∞

d. 100

40 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 7.02 Integration by Parts

General solution of $\frac{dy}{dx} = 2x$:

a. $y = x^2$

b. $y = 2$

c. $y = x^2 + C$

d. $y = 2x + C$

41 [Challenging] • Target: Another applied separable equation • Cross-Lesson: 8.01 Area Between Curves

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$

c. $y = 2x + 2$

d. $y = xe^x$

42 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 8.02 Volumes of Revolution

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$

43 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Separable ODE $\frac{dy}{dx} = \frac{y \ln y}{x}$, solution:

a. $\ln y = Cx$

b. $y = Ae^{Bx}$

c. $y = e^{Cx}$

d. $\ln(\ln y) = \ln x + C \Rightarrow \ln y = Ax$

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44 [Challenging] • Target: A final applied separable model • Cross-Lesson: 1.01 Precalculus Review for Calculus

Separable ODE: $\frac{dy}{dx} = \frac{y}{x}$. Solution:

a. $y = x + C$

b. $y = Ce^{x}$

c. $y = Cx$

d. $y = C/x$

45 [Challenging] • Target: General Solutions of Separable Equations • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

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